

DSpecM1D–QSSP waveform and performance comparison

Public-code benchmark on `earth-tunya`, 18 August 2026

Purpose. This small comparison asks whether two independent public implementations produce reasonably similar phase timing and broad waveform character when configured as closely as their user interfaces permit. It is not a numerical validation and neither code is treated as ground truth.

Public codes. DSpecM1D v1.0.0 (26 April 2026; commit 9b13dc56) was obtained from <https://github.com/adcm2/DSpecM1D>. QSSP 2024 was obtained from https://github.com/RongjiangWang/QSSP_2024 at public commit 907706b7 (4 July 2026; no release tag supplied). Both built successfully with GCC/G++/GFortran 15.2.0. DSpecM1D’s exact dependencies were checked out locally and supplied through standard CMake FetchContent overrides because the release combines old commit pins with depth-one clones; no source was edited.

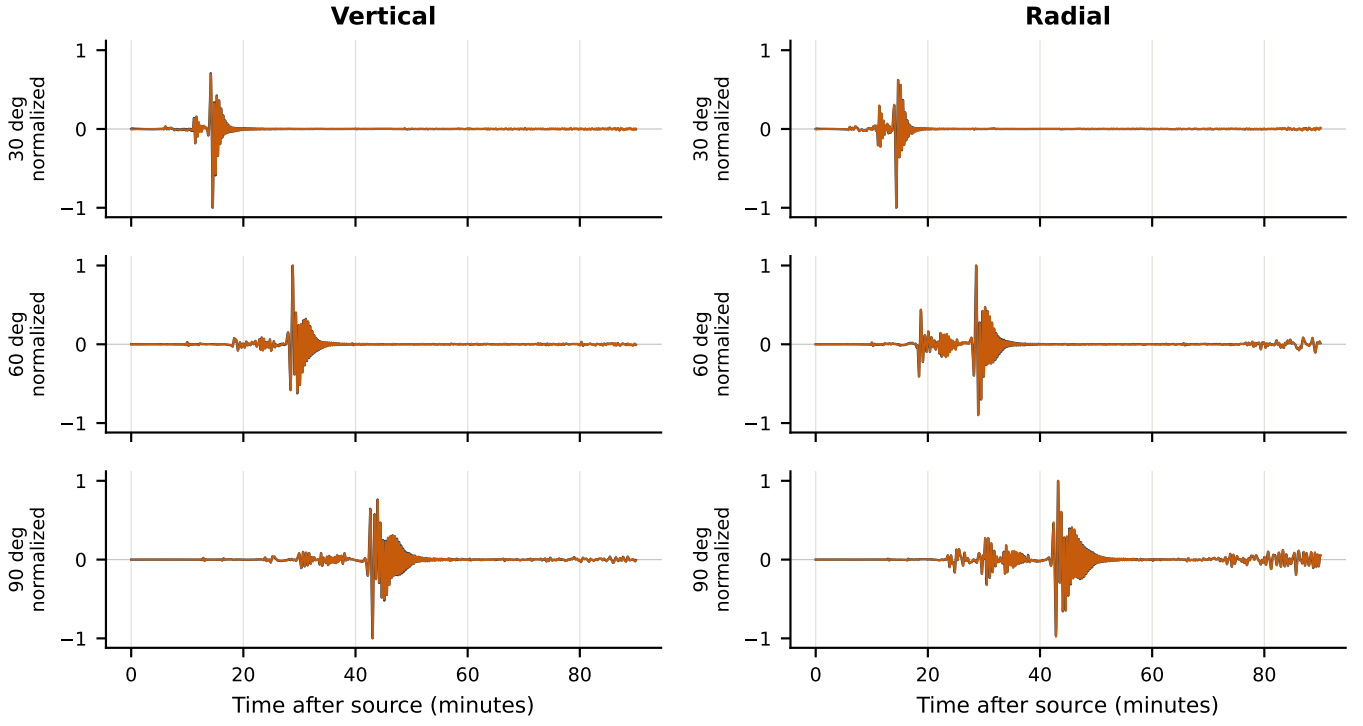
Machine. Two AMD EPYC 9334 processors (64 physical/128 logical cores), 1.48 TiB RAM, Linux 6.1, FFTW 3.3.10. Practical DSpecM1D runs used 50 OpenMP threads; QSSP was the documented serial Fortran build.

Property	DSpecM1D	QSSP
Earth model	Public <code>prem.200.isotropic</code> ; 220 radius samples	Same 220 samples reversed into depth-ordered layers
Material	Isotropic ρ, V_P, V_S ; solid surface; no ocean/atmosphere	Same samples and solid surface
Attenuation	Disabled by interface switch	Dispersion off; $Q_P = Q_S = 10^9$ in solids
Gravity/rotation	Full self-gravity; non-rotating	Gravity through full band/degree range; non-rotating
Boundary/receivers	Spherical free surface; receivers at 0 km depth	Same
Source	Explicit (r, θ, ϕ) moment tensor	Same tensor; normalized 1 s $\sin^2$ moment-rate pulse
Output	Displacement, Z, N, E	Displacement, E, N, Z files; Z upward
Plot treatment	Identical zero-phase external bandpass; signs retained; paired traces independently normalized	

Case	Source	Receivers	Band / record	Purpose
A	isotropic, 25 km	30, 60, 90°	2–45 mHz / 8191 s	body and first surface-wave trains
B	isotropic, 220 km	135, 145, 155°	2–35 mHz / 4095 s	PKP-family/core wavefield
C	DC 30/60/90, 25 km	90, 116°	2–18 mHz / 16382 s	long-period three-component global propagation

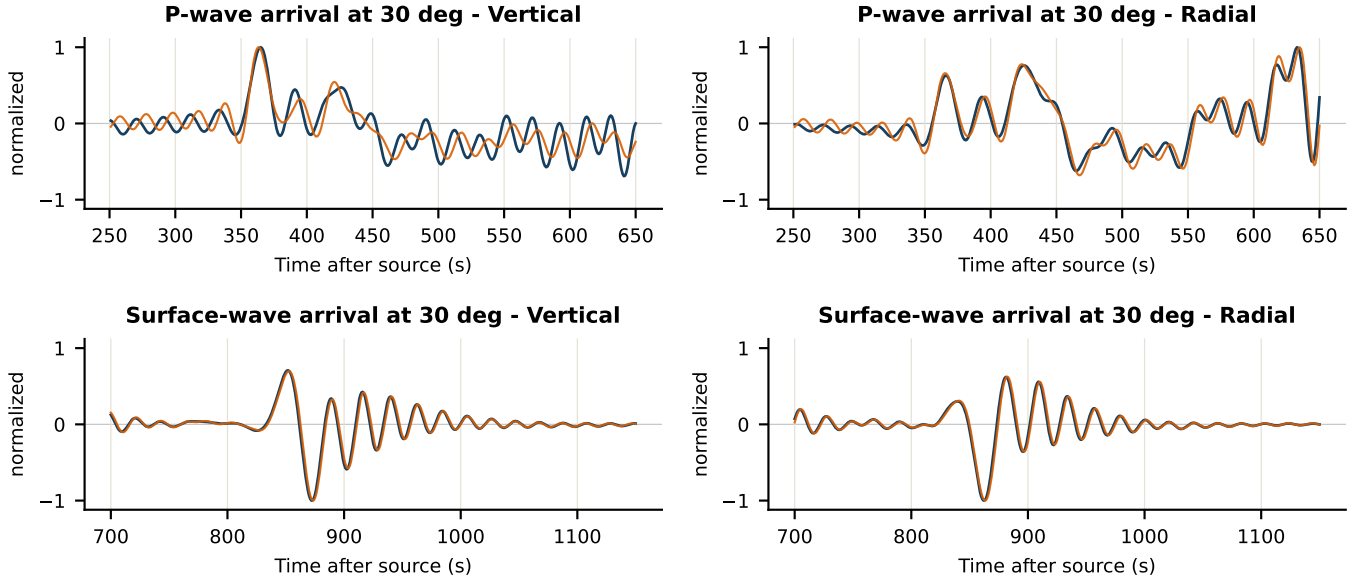
Benchmark A: independently normalized displacement

— DSpecM1D — QSSP



Common zero-phase bandpass: 2-45 mHz; signs retained

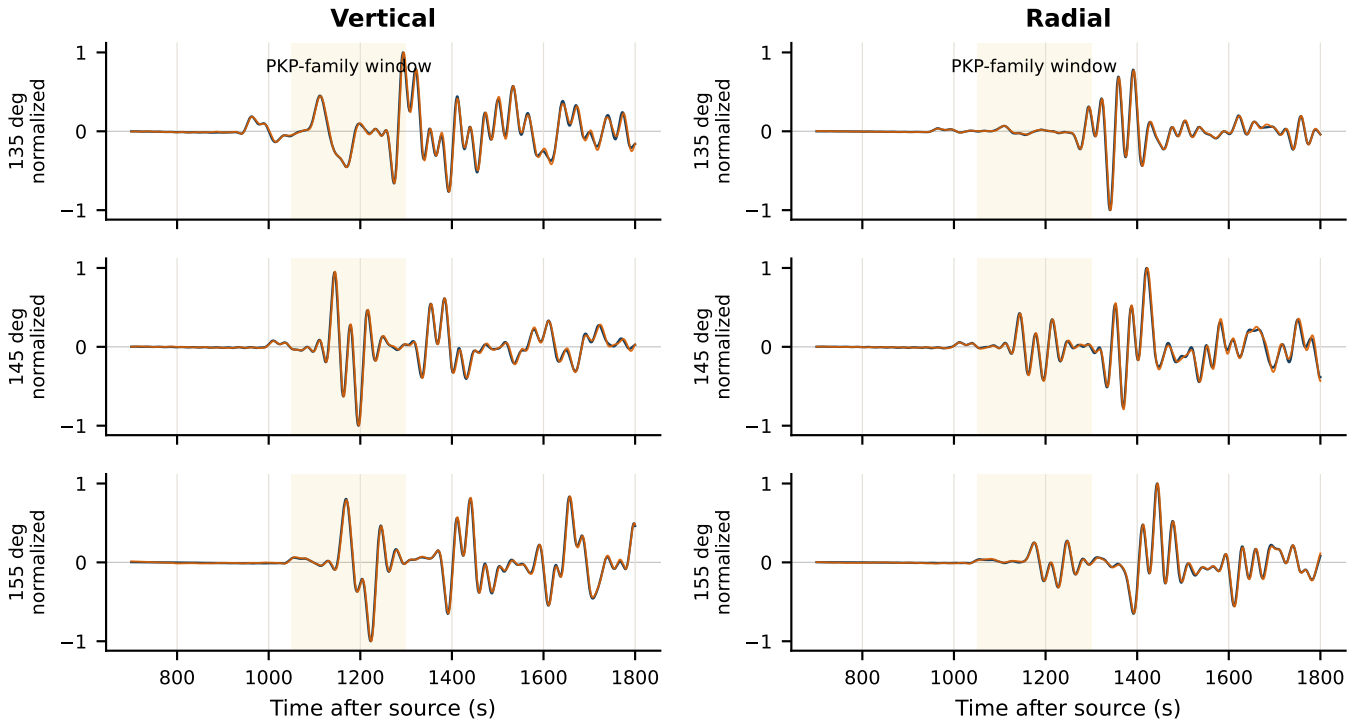
— DSpecM1D — QSSP



The major body-wave and surface-wave packets occur at the same times and retain the same polarity at all three distances. The 30° arrival-scale panels show approximately 5–10 dominant cycles around the P onset and surface-wave train; both remain closely superposed. Modest differences are confined mainly to low-amplitude coda.

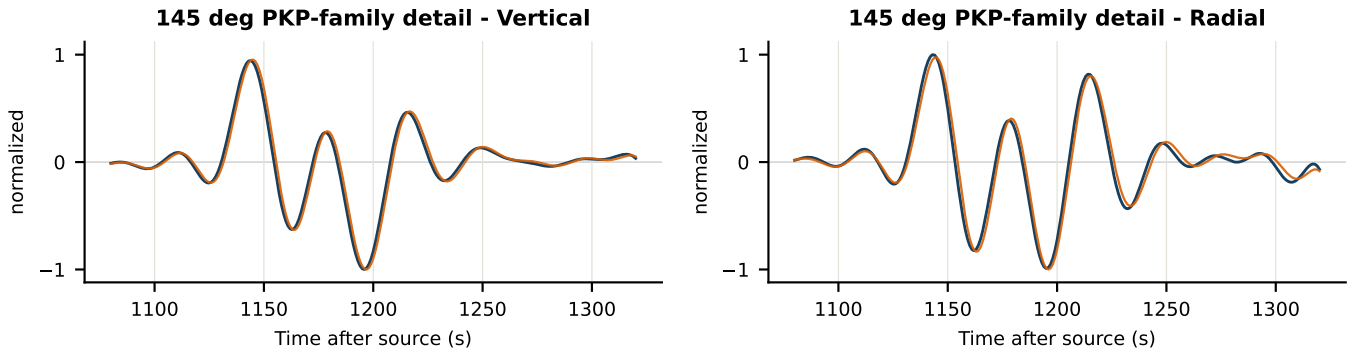
Benchmark B: independently normalized displacement

— DSpecM1D — QSSP



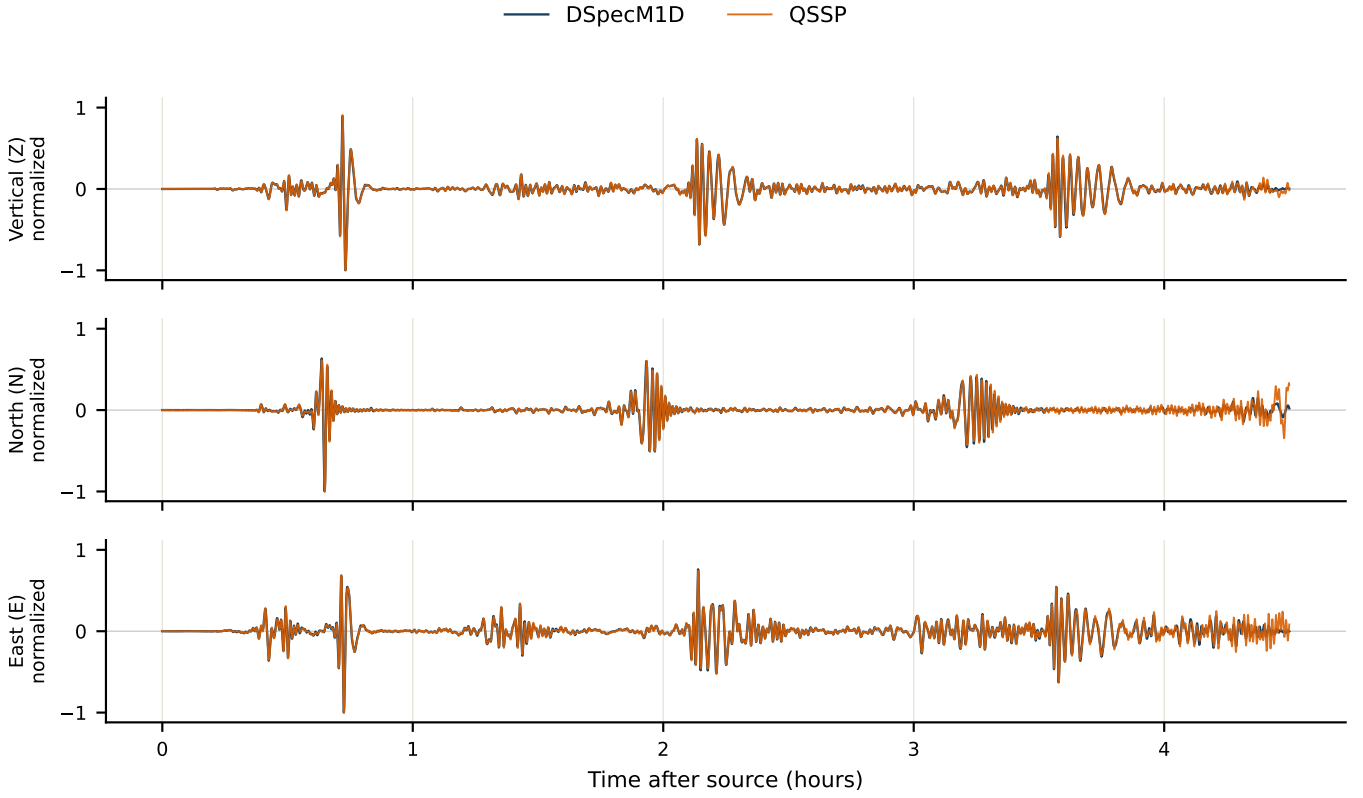
Common zero-phase bandpass: 2-35 mHz; signs retained

— DSpecM1D — QSSP

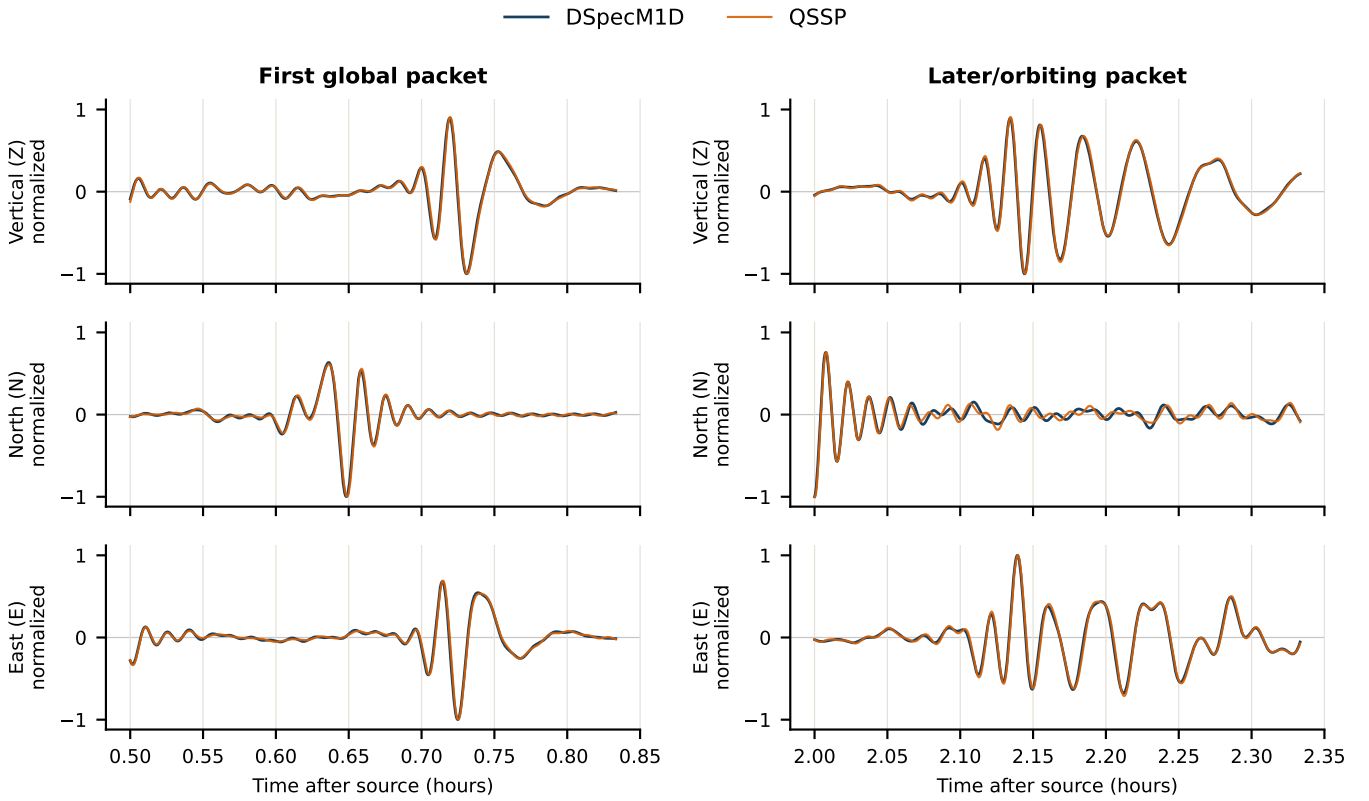


Across 135°, 145°, and 155°, the main core-sensitive arrivals align closely in phase and waveform character. The tighter 145° panel resolves individual oscillations within the PKP-family packet without assigning branches. The practical 40 mHz cap kept serial QSSP preprocessing finite; a late P4KP extension was not included.

Benchmark C: long-period three-component displacement at 90 deg



Common zero-phase bandpass: 2-18 mHz; each paired trace normalized independently; signs retained



At the 90° receiver, all three components show close timing and waveform agreement through the initial and later packets. The expanded 0.5–0.83 h and 2.0–2.33 h windows show the agreement cycle by cycle. Both spheroidal and toroidal motion are exercised; small low-amplitude differences remain unsurprising for independent algorithms.

Performance and summary

Timings are measured runs, not estimates. Peak RSS is Linux child-process `ru_maxrss`. QSSP “build” rows include Green spectra plus synthesis; its A reuse row reads saved spectra and synthesizes three receivers. DSpecM1D directly computes the requested spectra and time series.

Case	Code/stage	Threads	$f_{\max}$	Record	Wall (s)	Peak RSS (MiB)
A	DSpecM1D direct	50	50 mHz	8191 s	4.27	481.0
A	QSSP GF + synthesis	1	50 mHz	8191 s	1490.81	64.4
A	QSSP synthesis reuse	1	50 mHz	8191 s	0.57	13.4
B	DSpecM1D direct	1	40 mHz	4095 s	74.06	20.2
B	DSpecM1D direct	50	40 mHz	4095 s	1.58	375.6
B	QSSP GF + synthesis	1	40 mHz	4095 s	445.78	50.0
C	DSpecM1D direct	50	20 mHz	16382 s	1.25	152.3
C	QSSP GF + synthesis	1	20 mHz	16382 s	327.44	27.9

Interpretation. For all three configurations, major arrivals occur at similar times and the broad waveform character is consistent between the implementations. Benchmark B in particular gives a close visual overlay of the core-phase wavefield near 145° ; Benchmark C maintains that agreement across several hours and three components.

Caveats. Traces are independently normalized, so the figures do not test absolute source normalization. DSpecM1D uses spectral elements over the supplied radius samples, whereas QSSP consumes them as isotropic depth layers and uses a hybrid normal-mode/propagator algorithm. The one-second QSSP STF and narrow DSpecM1D edge taper also differ slightly. These distinctions make small residual waveform differences expected and do not alter the qualitative result.

Conclusion. Configured through their public interfaces with matched PREM samples and simple common physics, DSpecM1D and QSSP seismograms overlay closely in phase and overall waveform character. This supports practical consistency for ordinary teleseismic, core-phase, and long-period global calculations, without constituting formal numerical validation.